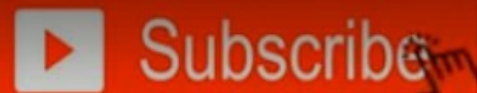


MTH622: Quiz 2



Question # 3 of 10

SOLVED BY: **WAQAR HUSSAIN**

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass m with velocity $3\hat{i}$. If after collision the velocity of first particle is $5\hat{i}$ and that of second particle is $2\hat{i}$. Find m

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Select the correct option



3



4





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SOLVED QUESTIONS



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Q.No.1: Write necessary and sufficient conditions for a Conservative Force Field.

(Module 97, Pg.34)

ANSWER: Conservative force fields conserve the following properties:

1- A force field $\mathbf{F}$ is conservative if and only if there exists a continuously differentiable scalar field V such that $\vec{\mathbf{F}} = -\nabla V$ or, equivalently, if and only if

$$\text{curl } \vec{\mathbf{F}} = \nabla \times \vec{\mathbf{F}} = \mathbf{0}$$

identically

2- A continuously differentiable force field $\mathbf{F}$ is conservative if and only if for any closed non-intersecting curve C (simple closed curve)

$$W = \oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = 0$$

i.e. the total work done in moving a particle around any closed path is zero.



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SOLVED QUESTIONS



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Q.No.2: Write different examples of Conservative Forces.

(Module 97, Pg.35)

ANSWER: Examples of Conservative Forces are given below:

- 1-** Gravitational force is an example of a conservative force.
- 2-** Elastic spring force is example of conservative force.
- 3-** The work done of a particle moving along a closed path is zero and the force which causes such motion is conservative.

Q.No.3: State the Chasles' Theorem.

(Module 109, Pg.23)

ANSWER: Theorem Statement:

Chasle's theorem states that the most general rigid body displacement can be produced by a translation along a line (called its screw axis) followed (or preceded) by a rotation about that line.



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SOLVED QUESTIONS



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Q.No.4: Write down the mathematical expression for angular momentum of a single particle.
(Module 135, Pg.64)

ANSWER: The angular momentum of a single particle is defined as the cross product of linear momentum and position vector of concerned particle.
Mathematically
$$L = r \times mv$$

Q.No.5: State Euler's Theorem.
(Module 108, Pg.21)

ANSWER: Theorem Statement:
"A rotation of a rigid body about a fixed point of the body is equivalent to a rotation about a line which passes through the point."



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Q.No.6: Write down examples of Curvilinear Motion.
(Module 112, Pg.27)

ANSWER: Examples of curvilinear motion are:

- 1- A stone thrown into the air at an angle.
- 2- A car driving along a curved road.
- 3- Throwing paper airplanes or paper darts is an example of curvilinear motion.



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SOLVED QUESTIONS



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Q.No.7: Show that $\vec{F} = xy\hat{i} + 2z\hat{k}$ is non-conservative force. (Module 101, Pg.10)

SOLUTION:

$$\vec{F} = xy\hat{i} + 2z\hat{k}$$

To verify whether the force field $\mathbf{F}$ is conservative or not, we will check whether $\nabla \times \vec{F} = \mathbf{0}$ or not.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & 0 & 2z \end{vmatrix}$$

$$= \hat{i} \left| \frac{\partial}{\partial y} (2z) - \frac{\partial}{\partial z} (0) \right| - \hat{j} \left| \frac{\partial}{\partial x} (2z) - \frac{\partial}{\partial z} (xy) \right| + \hat{k} \left| \frac{\partial}{\partial x} (0) - \frac{\partial}{\partial y} (xy) \right|$$

$$= (0 - 0)\hat{i}$$



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SOLVED QUESTIONS



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$$= (0 - 0)\hat{i} - (0 - 0)\hat{j} + (0 - x)\hat{k}$$
$$= -x\hat{k}$$



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SOLVED QUESTIONS



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$$= (0 - 0)\hat{i} - (0 - 0)\hat{j} + (0 - x)\hat{k}$$

$$= -x\hat{k}$$

As we obtain that $\nabla \times \vec{F} \neq \vec{0}$ so we
Conclude that $\vec{F}$ is non-conservative.

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Q.No.8: If $r = a \cos \omega t \hat{i} + b \sin \omega t \hat{j}$ and $p = -am\omega \sin \omega t \hat{i} + bm\omega \cos \omega t \hat{j}$
Then find the angular momentum. (Module 120, Pg.39)

SOLUTION:

$$r = a \cos \omega t \hat{i} + b \sin \omega t \hat{j}$$

$$p = -am\omega \sin \omega t \hat{i} + bm\omega \cos \omega t \hat{j}$$

As we know that

$$r \times p = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a \cos \omega t & b \sin \omega t & 0 \\ -am\omega \sin \omega t & bm\omega \cos \omega t & 0 \end{vmatrix}$$

$$= (abm\omega \cos^2 \omega t + abm\omega \sin^2 \omega t) \hat{k}$$

$$= abm\omega (\cos^2 \omega t + \sin^2 \omega t) \hat{k}$$

$$= abm\omega (1) \hat{k}$$

$$= abm\omega \hat{k}$$

$$\tau = \frac{d\Omega}{dt} = \frac{d}{dt} (r \times p) = 0$$

We know that $\Omega = r \times p$

$$\Omega = abm\omega \hat{k}$$

It is the required angular momentum.



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Q.No.9: Particles of masses $2m$, $3m$ and $4m$ are held in a rigid light framework at points $(1, 0, 1)$, $(1, 1, 0)$ and $(0, 1, 1)$. Calculate moment of inertia of x-axis.

(Module 160, Pg.120)

SOLUTION:

Given that:

$$m_1 = 2, \quad m_2 = 3, \quad m_3 = 4$$

Points are

$$P_1(1, 0, 1), \quad P_2(1, 1, 0), \quad P_3(0, 1, 1)$$

Moment of Inertia about x-axis is

$$I_{xx} = \sum_i m_i (y_i^2 + z_i^2)$$

$$I_{xx} = m_1 (y_1^2 + z_1^2) + m_2 (y_2^2 + z_2^2) + m_3 (y_3^2 + z_3^2)$$

$$I_{xx} = 3(0 + 1) + 5(1 + 0) + 7(1 + 0)$$

$$I_{xx} = 3 + 5 + 7 \quad I_{xx} = 15$$

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Q.No.1: What is the impulse developed by a mass of **100 gm** which changes its velocity from **$i + j + k$ m/sec** to **$2i + 2j + 3k$ m/sec**? (Module 118, Pg.37)

SOLUTION:

Mass **$m = 100$ gm**

Initial velocity
 $v_1 = i + j + k$

Final velocity
 $v_2 = 2i + 2j + 3k$

Impulse = **$m(v_2 - v_1)$**

= **$100((2i + 2j + 3k) - (i + j + k))$**

= **$100(2i + 2j + 3k - i - j - k)$**

= **$100(i + j + 2k)$**

Impulse Magnitude

= **$100\sqrt{(1)^2 + (1)^2 + (2)^2}$**

Impulse Magnitude

= **$100\sqrt{1 + 1 + 4}$**

Impulse Magnitude

= **$100\sqrt{6}$**

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MTH022

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Q.No.2: If moment of Inertia of a uniform circular cylinder of length b and radius a about **x-axis** through the center and perpendicular to the central axis is:

$$I_x = m \left(\frac{1}{4} a^2 + \frac{1}{12} b^2 \right)$$

Find the moment of Inertia about **y-axis**?

(Module 168, Pg.138)

SOLUTION:

Moment of Inertia about x-axis $I_x = m \left(\frac{1}{4} a^2 + \frac{1}{12} b^2 \right)$

Due to symmetry we have $I_x = I_y$

Moment of Inertia about y-axis will be $I_y = m \left(\frac{1}{4} a^2 + \frac{1}{12} b^2 \right)$



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Q.No.3: Write the expression for total energy of a particle executing Simple Harmonic Motion.
(Module 103, Pg.13)

ANSWER: If T is the kinetic energy and V is the potential energy then total energy will be $E = T + V$ where

$$K.E = T = \frac{1}{2}mv^2$$

and $V = \frac{1}{2}kx^2$

Then the total energy of S.H.M will be

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

Q.No.4: Consider a small element of a uniform elliptic plate has moment of inertia $I = \delta m y^2$. Write down the expression for moment of inertia of whole plate.
(Note: Do not solve the expression).
(Module 172, Pg.146)

ANSWER: The moment of inertia of this element along x-axis will be equal to $I = \delta m y^2$. Then the moment of inertia of whole plate will be

$$I_{xx} = \int (\delta m) y^2 = \int_{\text{plate}} \rho ds y^2$$



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Q.No.5: To find the inertia matrix of uniform solid cuboid (parallelepiped) at one of its corner. If $I_{xx} = \frac{M}{3}(b^2 + c^2)$, then find I_{yy} and I_{zz} . (Module 157, Pg.110)

SOLUTION:

$$I_{xx} = \frac{M}{3}(b^2 + c^2)$$

$$I_{yy} = \frac{M}{3}(a^2 + c^2)$$

Similarly, due to symmetry, we can write

$$I_{zz} = \frac{M}{3}(a^2 + b^2)$$

Q.No.6: What is the degrees of freedom of the system of rigid bodies. (Module 109, 122, Pg.21,42)

ANSWER: "The number of coordinates required to specify the position of a system of one or more particles is called the number of degrees of freedom of the system."

A rigid body has **six** degrees of freedom.



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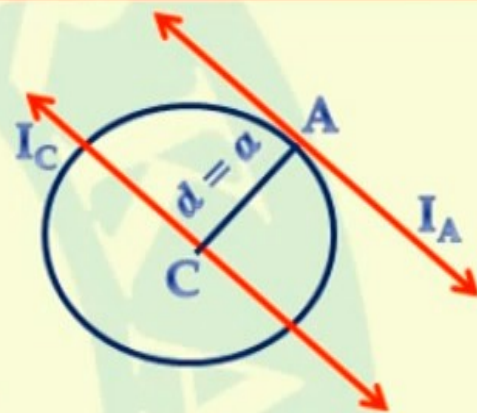
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Q.No.7: Use the parallel axis theorem to find the moment of inertia of a solid circular cylinder about a line on the surface of the cylinder and parallel to axis of cylinder.

(Module 120, Pg.39)



SOLUTION: By parallel axis theorem

$$I_A = I_C + Ma^2 \dots \dots (1)$$

Since I_C which is the moment of inertia of a solid circular cylinder about an axis passing from the center of mass is defined by

$$I_C = \frac{1}{2} Ma^2 \dots \dots (2)$$

By substituting equation (2) in equation (1), we have

$$I_A = \frac{1}{2} Ma^2 + Ma^2$$

Taking Ma^2 common, we get

$$I_A = Ma^2 \left(\frac{1}{2} + 1 \right)$$

$$I_A = \frac{3}{2} Ma^2$$

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MTH622: Quiz 2



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Question # 1 of 10

SOLVED BY: **WAQAR HUSSAIN**

If the axis of rotation passes through the center of mass of the rigid body then it is called

Select the correct option

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revolution



spin

If the **axis of rotation** passes through the center of mass of the rigid body then body is said to be **spin** or **rotate** upon itself.



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Question # 1 of 10

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If the axis of rotation passes through the center of mass of the rigid body then it is called

Select the correct option

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revolution



spin

(Module 122, Pg.42)



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Question # 2 of 10

SOLVED BY: **WAQAR HUSSAIN**

If an object explodes, the center of mass will

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Select the correct option



continue to move in the same direction with the same speed.



continue to move in the same direction with a different speed.



If an object **explodes**, the different pieces of the object will follow seemingly independent paths after the **explosion**. The **center of mass**, however, will **keep doing** what it was doing **before the explosion**. This is because an explosion involves only **internal forces**.





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Question # 2 of 10

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If an object explodes, the center of mass will

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Select the correct option



continue its previous motion

(Module 124, Pg.45)



come into rest



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